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M. TECH.
(SEM II)THEORY EXAMINATION 2018-19
ADVANCED OPTICAL NETWORKS

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief. 2 x 7 = 14**

- a. What are the types of WDM and mention their advantages
- b. What are the types of MPLS nodes?
- c. What is path protection switching?
- d. Define Point to point topology.
- e. Mention the importance of multiplexing.
- f. List the applications of tunable DWDM Lasers.
- g. What is the function of line protection and path protection switching?

SECTION B**2. Attempt any three of the following: 7 x 3 = 21**

- a. Explain the principle of operation of WDM in optical communication with its advantages and disadvantages.
- b. Explain the functionalities of MPLS nodes.
- c. Summarize the relationship between OXC and MPLS Operations with the help of a diagram.
- d. What are the various types of Network Topologies? Explain
- e. Explain about the benefits of using SONET/SDH compared to other networks.

SECTION C**3. Attempt any one part of the following: 7 x 1 = 7**

- (a) Explain the different multiplexing techniques in optical networks.
- (b) Organize the details of MPLS Traffic Engineering and traffic oriented performance.

4. Attempt any one part of the following: 7 x 1 = 7

- (a) Explain the advantages of switching in optical networks and how IP forwarding differs from label switching in optical networks?
- (b) Illustrate the role of Optical Add/Drop Multiplexer (OADM) in providing cost effective means of handling traffic in optical networks.

5. Attempt any one part of the following: 7 x 1 = 7

- (a) Briefly explain the SDH frame structure and show that the basic bit rate is 51.84 Mbps.
- (b) Explain the basic SONET transmission format with the aid of a diagram.

6. Attempt any one part of the following: 7 x 1 = 7

- (a) Explain Link Management protocol (LMP).
- (b) Explain Optical compilers. What is Time multiplex multiprocessor?

7. Attempt any one part of the following: 7 x 1 = 7

- (a) Mention the details of SONET and SDH Multiplexing hierarchy with a detailed diagram.
- (b) Explain the different kinds of protection switching.